Shao-Yu Huang

Hsinchu City, Taiwan | 0963809732 | sandyhsy0447@gmail.com

EDUCATION

San José State University — San José, California
Master of Science, Applied Data Intelligence — May 2026

National University of Kaohsiung — Kaohsiung, Taiwan
Bachelor of Information Management — April 2024

EXPERIENCE

San José State University — Oct 2025 - May 2026
Graduate Research Assistant, NEAR Lab | Attention-Aware Eye-Tracking — San José, California

- Developed an end-to-end multimodal AI pipeline integrating eye tracking, computer vision, gaze mapping, and vision-language models (GPT-4.1) to transform user attention into grounded, explainable behavioral reasoning.
- Engineered gaze-processing workflows for AOI extraction, Gaussian heatmap generation, synchronized video visualization, and pixel-level gaze alignment to support multimodal reasoning and representation learning.
- Built an attention-aware inference framework that grounded LLM responses with gaze-derived visual context, enabling interpretable natural-language explanations for human-AI interaction.

PROJECTS

Multimodal RAG for Document Question Answering | *RAG, Sentence Transformers, LLM*

- Built an end-to-end Retrieval-Augmented Generation pipeline using SentenceTransformers, Gemini, and multimodal PDF parsing to retrieve grounded evidence from text, figures, and tables.
- Designed a dual-index semantic retrieval architecture combining textual and visual embeddings, achieving 0.46 Kaggle score while reducing hallucinations through evidence-grounded generation.

Multimodal Fashion Retrieval and Representation Learning | *CLIP, VLM Fine-Tuning*

- Developed the FACTOR multimodal fashion recommendation framework using CLIP fine-tuning and representation learning, improving average attribute prediction by 12% over CLIP and FashionCLIP baselines across 261K fashion samples.
- Proposed a contrastive learning-based semantic evaluation framework using CLIP embeddings to measure semantic correctness beyond exact-label matching and improve model evaluation.
- Built an attribute-aware semantic retrieval pipeline leveraging structured garment representations and embedding similarity to improve fine-grained fashion search under ambiguous user queries.

Multimodal Vision-Language Understanding | *CLIP, BERT, LLaVA, Qwen, VLM, Fine-Tuning*

- Developed a multimodal vision-language framework integrating CLIP, BERT, LLaVA/Qwen-generated captions, and transformer-based cross-attention for 14 abstract fashion style recognition across 14 categories.
- Designed a cross-attention fusion mechanism to align visual and textual representations, improving multimodal representation learning for fine-grained style understanding.
- Achieved 0.85 Macro F1 through CLIP fine-tuning and multimodal fusion, outperforming the original benchmark by 26 percentage points.

SKILLS

AI / LLM: RAG prompt engineering, LLM API integration, embeddings, semantic retrieval, structured generation

Languages / Full Stack: Python, React, Node.js, Express, FastAPI, SQL

Data / Databases: MySQL, MongoDB, SQLite, Redis, FAISS, ETL/ELT, star schema

Cloud / Distributed Systems: AWS (EKS, ECR, S3), Docker, Kafka, Apache Spark, REST APIs

ML: PyTorch, CLIP, BERT, SentenceTransformers, MobileNetV2, ResNet50, cross-attention, fine-tuning

RESEARCH & PUBLICATIONS

- S. Tang, A. Shah, A. S. Gawai, S.-Y. Huang, G. Liu, J. Xie, G. Huang. “Evaluating an AI-enhanced power wheelchair obstacle alert system: Effects of multimodal warning design on warning-response behavior.” Transportation Research Part F: Traffic Psychology and Behaviour, 123, 103801, 2026. DOI: 10.1016/j.trf.2026.103801.
- D. Han, S.-Y. Huang, J. Liang, M. Masum. “Multimodal StyleFusion: Cross-Attention Learning for Fashion Style Recognition.” Machine Learning with Applications, 25, 100961, 2026. DOI: 10.1016/j.mlwa.2026.100961.